

FINAL YEAR PROJECT 2024-2025

Ukulele Robot with Mobile Application

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Abstract

The project aims to create an ukulele robot that can interact with a mobile application. The robot will be able to play the note or chord on remote. The application can also act as accompaniment, to let the user have an real-time interactive experience.

Index Terms

Ukulele, Robot, Mobile Application

CONTENTS

I	Introduction	3
II	Literature Review	4
III	Methodology	5
III-A	Sequence Diagram	5
III-B	Flow Chart	6
IV	Current Progress	8
IV-A	Hardware	8
IV-A1	Electronic Component	9
IV-A2	3D Print	13
IV-B	Software	20
IV-B1	Introduction	20
IV-B2	Mobile Application	20
IV-B3	Raspberry Pi	22
IV-B4	Arduino Uno	24
V	Next Steps	25
V-A	Hardware	25
V-B	Software	25
	References	27

I. INTRODUCTION

When learning string instrument as a beginner, such as guitar and ukulele, it is common to search for different chord and sometime it may take a long time to find all the necessary chords and fingering styling. Also sometime, beginner would like a demo of how the song is play as a reference to check for own understanding and performance. [1] Concerning another situation that the sound track of a song you are interested in cannot be found online, but you would like to listen to it now, or you would like to preform duet but you have no partner. To address all those problem, a solution is proposed with seek to merge technology and music by providing solutions to both music lover and music learners.

The merging of a robotic system with a mobile App focuses on increasing engagement when playing the ukulele while also providing a solution to the growing demand for automated Instruments. The ability to play difficult chords and techniques means that the ukulele robot allow the average user to enjoy the experiences of live performances without the performer having to be particularly skilled; this makes music easy and accessible even for novice players. Moreover, this project also addresses the issue of music education as it provides an interesting instrument for practice and learning with which the users can learn various songs and styles at their own pace. All in all, this initiative promotes creativity and discovery into music, inculcating the users with a broader sense of understanding and appreciation of musical arts and the use of technology to achieve this.

This report outlines the approach of making a ukulele robot with an accompanying mobile application to create more engaging interactions with music. The main focus of the project is to obtain a robotic system, which is fully autonomous that can play a ukulele and enable the users to pick a song and change the chords in real-time through controlling. The hardware includes a multi-arm mechanism to preform string pressing and servo motors to wiggle the strings by mimicking various fingering techniques. The mobile app allows the users to select songs to play with, therefore improving the overall experience of the user. Early tests confirm that almost all chords were played by the robot with hard strokes but little sound which is one of the problems existing in all robotic musical instruments. This project is not only related to robotic musical performances but also probes new ways for making music and interaction with the audience.

II. LITERATURE REVIEW

Integrating robotics into the music industry has become increasingly common in recent years. A robotic musical instrument is characterized as a device that autonomously generates music through the utilization of mechanical components, including motors, gears, and solenoids [2]. With the incorporation of technology, this facilitates performance, accessibility, and even composition of music. The convergence of robotics and musical art has led to the development of diverse types of musical robots, including piano robots, turntable robots, percussion robots, string robots, and wind robots [2].

After studying the latest trends in the area of string robots, especially the guitar and ukulele robots, it was found that although there are many recorded resources on guitar projects, there is less documented information on ukulele robots, making it challenging for users to learn about them. Additionally, most of the current ukulele projects incorporate solenoid mechanisms which attract and release a magnet for string pressing[3].

After reviewing multiple solution online, it has been observed that the approach of using solenoid will result in creating significant noise that affect the user experience as it interrupted the musical performance [4]. While a minor of current project address such unwanted sound from happening by replacing each solenoid with servomotor and a arm structure to press the string. However, the approach of this project only focus on providing limited pressing of the chord as it only install the servomotor in few frets but not allowing user to access and press all the chord they can [5].

While the above research project are all using computer to directly control the robot, it has been find out another project that integrate a screen with the robot to enable control of the robot directly without bring computer all along [6].

After understand the different approach of the current project in the market and their pros and cons, we would like build a project that can provide better user experience in terms of quality of the music, user-friendliness and convenience.

III. METHODOLOGY

This project is composed of the collaboration of both hardware and software.

The hardware part include multiple parts to simulate the human interaction with the Ukulele. First, the robot is equipped with multiple arms. The arms are connected to a slider to control the vertical move and a extra motor for horizontal move. The arms are going to precisely press chords and perform different fingering techniques. This design is to simulate the human playing.

The software is an additional to the hardware component. The mobile application will allow the user to choose their own songs. It can also let the user to play a specific chord or fingering technique. It can provide an interface to the users on using the self-playing function of the robot. Integrating with software, the robot can improve the performance of playing music and provide personalized music experience.

Here are our system design diagrams:

A. Sequence Diagram

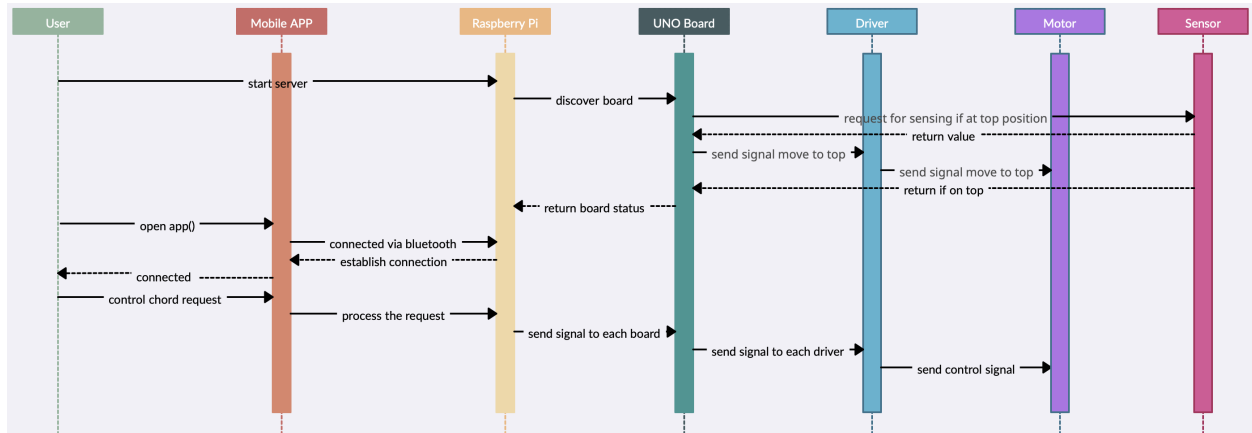


Fig. 1. Sequence diagram of the flow of whole system

Step-By-Step Descriptions:

- 1) Once The server started, the mechanical arms will be requested to reset to initial position by Uno.
- 2) It will check with the sensor at the initial position to see if the arms are of right position.
- 3) If it is reported that the arms are not in the initial position, Uno will send signal to each driver to request them to move to top.

- 4) Uno will send signal to Raspberry Pi notify the reset setup is done to let it broadcast the Bluetooth LE Signal.
- 5) The user open the app.
- 6) The app would receive the Bluetooth LE signal from robot.
- 7) The mobile app would request to connected with Raspberry Pi via Bluetooth.
- 8) Raspberry Pi establish connection with the mobile app via Bluetooth.
- 9) User input the chord they want to play in the mobile app.
- 10) Mobile app send chord request or command to Raspberry Pi.
- 11) Raspberry Pi process the chord request send signal to corresponding UNO.
- 12) Each UNO after receiving request will send signal to corresponding motor and preform movement.

B. Flow Chart

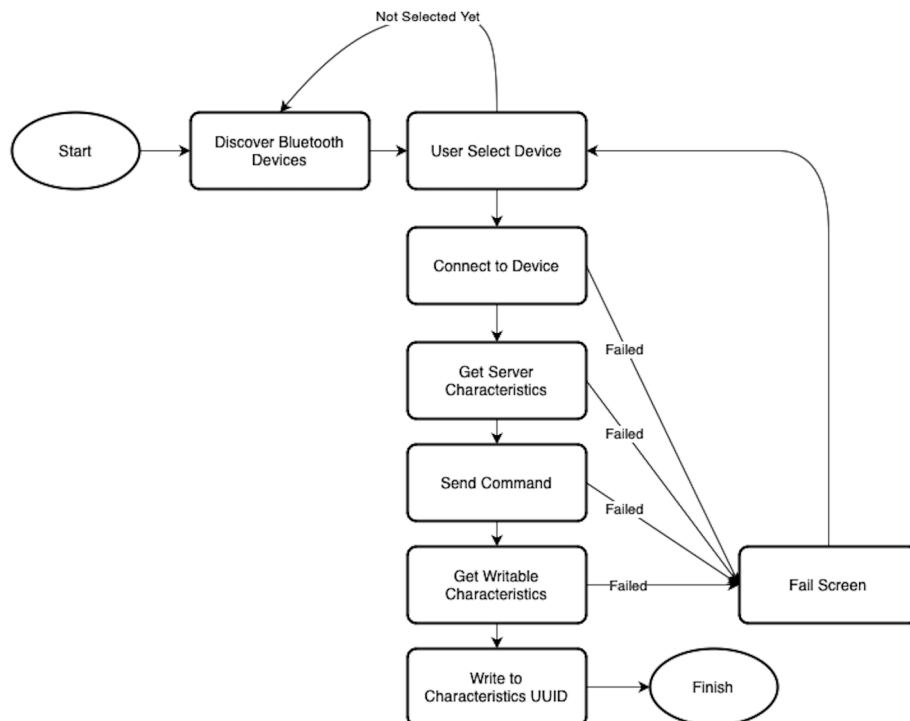


Fig. 2. Flowchart of Mobile Application.

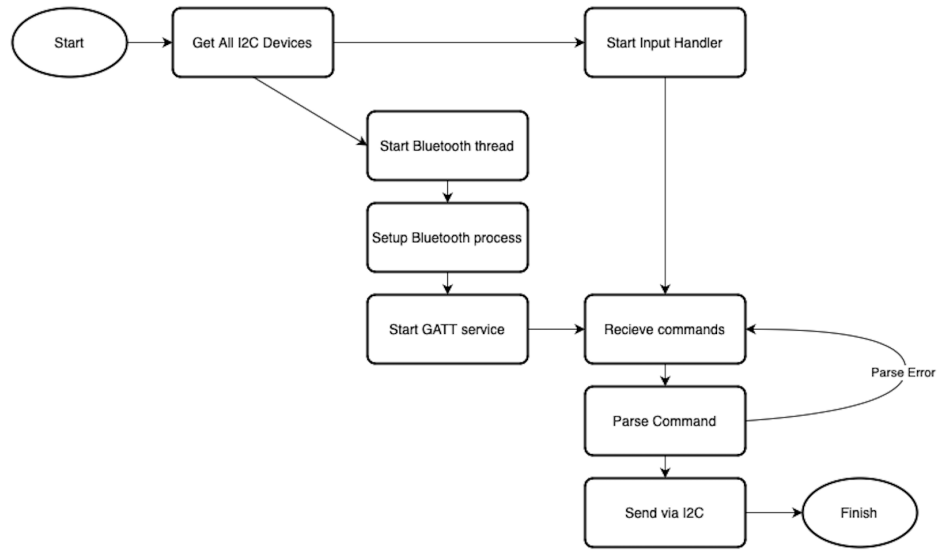


Fig. 3. Flowchart of Raspberry Pi.

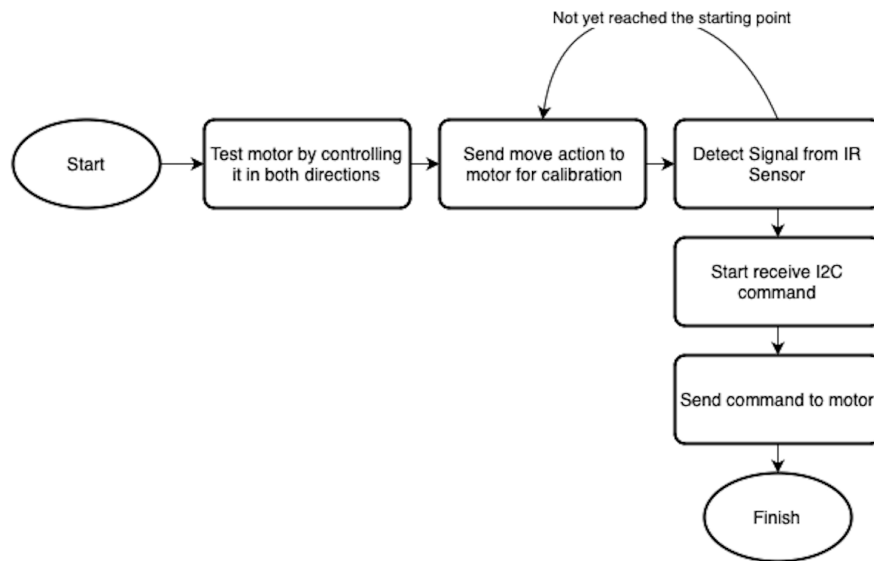


Fig. 4. Flowchart of Arduino Uno.

IV. CURRENT PROGRESS

A. Hardware

Key to this project is the ability to play the ukulele which is allowed by the arms being equipped with functional elements attachments and incorporating the slider mechanism into the project to provide better movement to the arm.

To ensure consistency in performance, the ukulele, sliders, and motor drivers will be fastened on an wooden board. This enhances effective mechanical interaction and ensures that there will not be disturbances while performing. Due to the defined placement of these constituents, the body natural movements are acted to ensure that the sound remains undisturbed after featuring machine interactions.

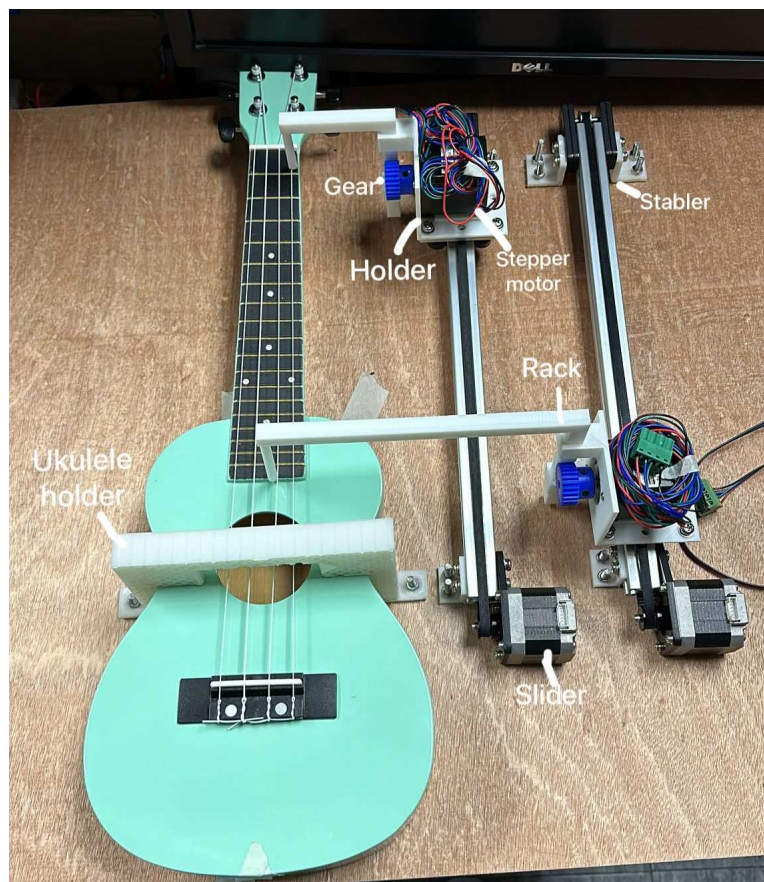


Fig. 5. Close up of the ukulele robot with each component labeled.

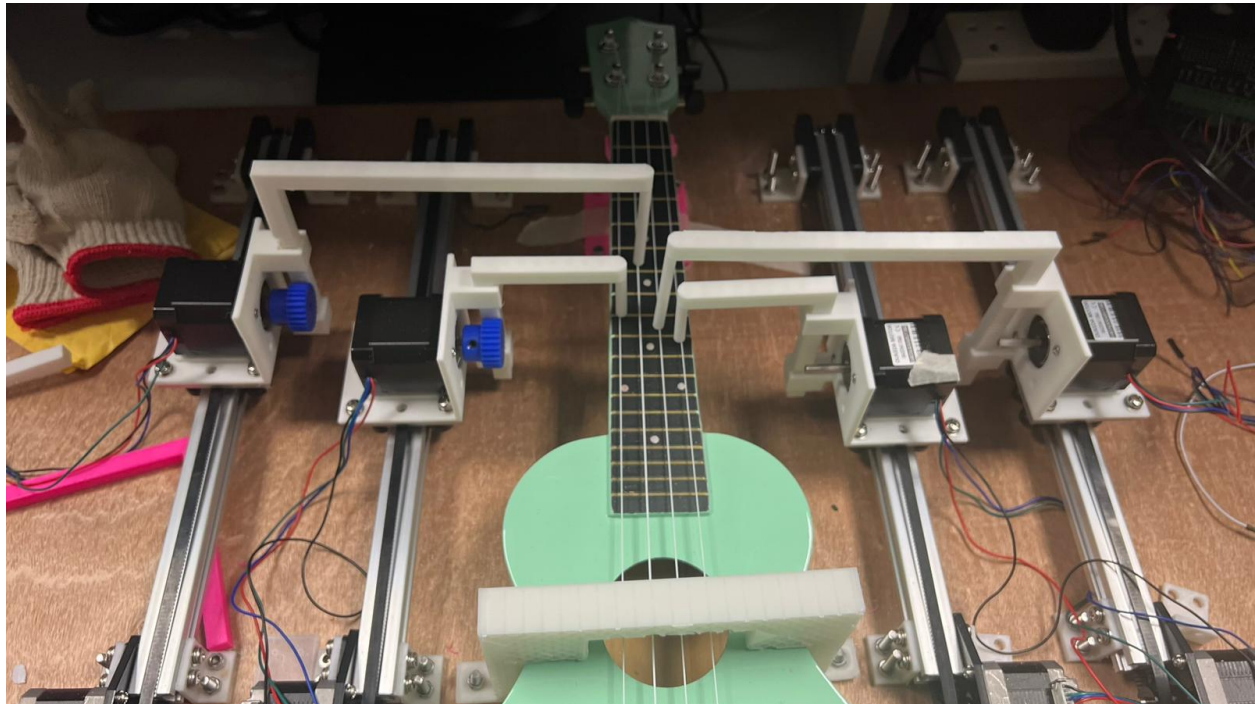


Fig. 6. Top view of the ukulele robot.

1) Electronic Component: In this project, different electronic components are embedded to the system to help bring the idea to practical. The type of electronic components need to be wisely chosen to preform the expected functionality. There are three main part we have to consider, namely, sliders, stepper motor and servomotor.

First, concerning the function of moving along the z-axis. It is consider to use a slider to preform high speed movement along the string, as the action of pressing different chords needed to be execute within a short period of time, almost instantly in order to catch up with the tempo of the performance. The speed and stability of the slider is the primary decision factor of choosing the appropriate slider. There are mainly two type of slider, ball screw driver and belt track slider. Ball screw slider make use of the recirculating ball mechanism between screw shaft and the nut to preform highly reliable horizontal movement, while it takes time to move from one position to another due to the underlying working rational. The accessible distance is comparatively short and may produce unwanted noise. While belt slider is capable of sliding on a longer strokes at a higher speed with less noise.

The reciprocating 42-step slider has been used to features belt transmission. This physical characteristic of the slider, which allows for linear movement within a short period of time, is important in ensuring that any unnecessary motion of the slider due to vibrations does not occur. Preventing deviation of the slider ensures consistent operation.

After handling the horizontal movement of the arm to locate the precise position of each fret, the next concern is the implementation of the vertical movement of the arm.

The common approach for such string-pressing functionality is performed using solenoids placed on top of every intersection of the frets and strings, pressing the string by releasing the armature onto it to apply pressure, mimicking the pressing action of a human finger. This approach requires a large number of solenoids, hence a significant amount of wire for connection [7]. Another, crucial issue mentioned in the literature review is the unavoidable clicking noise when the armature is released by the solenoid.

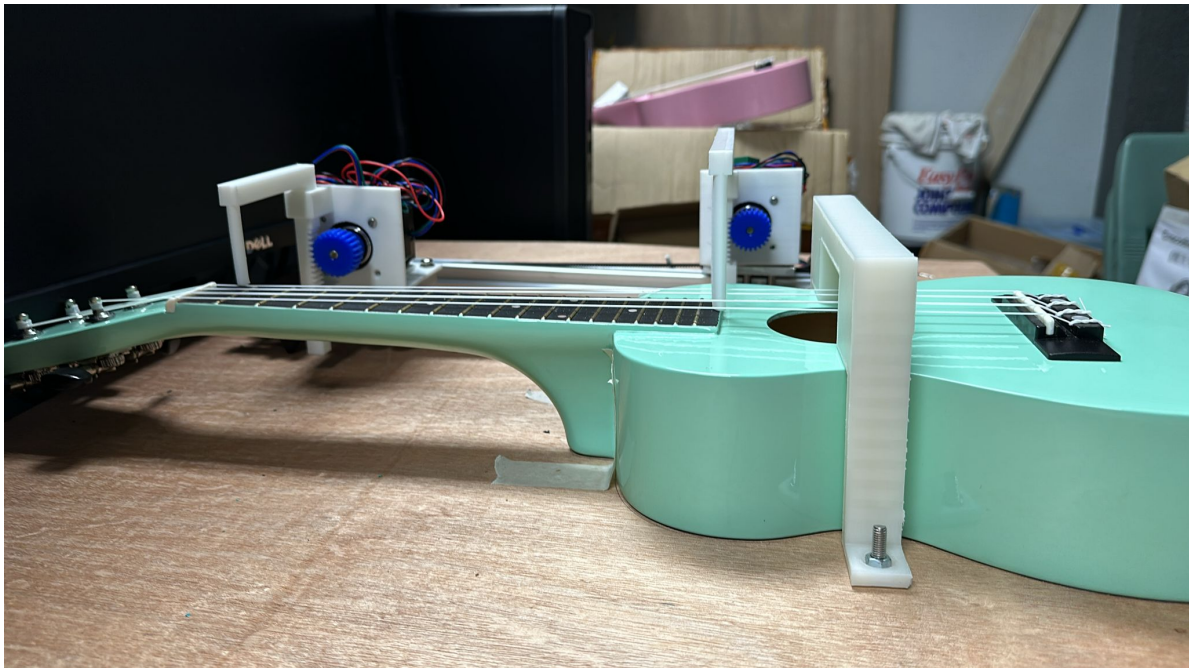


Fig. 7. Side view of the ukulele robot showcasing the structure of the gear and rack.

To prevent such noise disruptions that affect user experience and enjoyment, our project decided to apply a stepper motor incorporated with a gear and rack to manipulate the vertical action of string pressing. Stepper motors allow precise control of the angle of rotation, and thus the movement of the rack.

There are typically three types of stepper motor namely, permanent magnet stepper motor, variable reluctance stepper motor and hybrid stepper motor, which is the combination of the previous two. Permanent magnet stepper motor make use of the permanent magnet to control the angular movement by creating two magnetic field to interact with one another providing high torque while maintaining low speed. Variable reluctance stepper motor make use of the interaction between magnetic fields and soft iron rotor to perform rotation offering comparatively low cogging torque. Where hybrid stepper motor combine both structure resulting in the benefit of aforementioned stepper motor[8]. Therefore, hybrid stepper motor has been choose for the project.

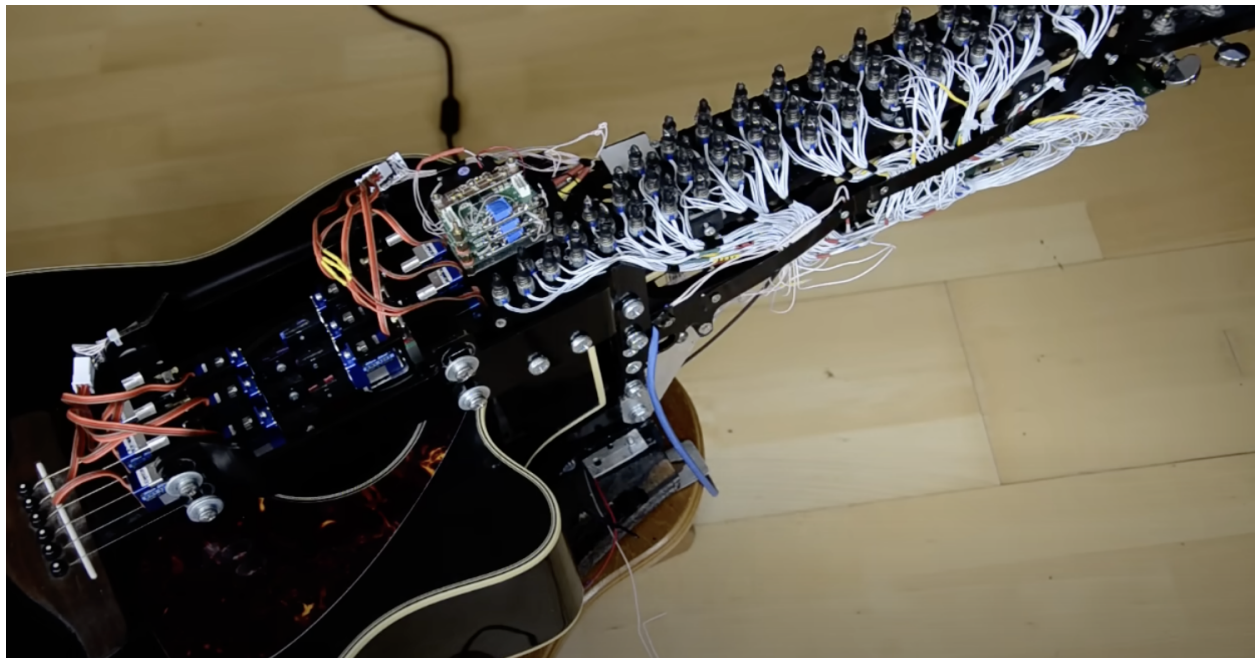


Fig. 8. Common practice of using solenoids with a bunch of wire needed.

Among different types of hybrid stepper motor, NEMA 17 stepper motors is famous for the

its high torque and reliability which matches with the goals of our project. While stepper motor only support the rotation, a mechanism is needed to design a transform from rotation to vertical movement, which a gear install onto the shaft with a rack that will move correspondingly when the gear is rotated. On top of each slider, a NEMA 17 stepper motor is placed with a geared rack, providing movement along the z-axis to ensure that the robot can form proper chord shapes on the strings.

Facilitate the calibration of the system, an infrared sensor is position at the top of the slider to check if the block is at the top.

For the sweeping, servomotor is consider to be use along with blade which allow the precise control of angular rotation. Unlike stepper motor, servomotor can be of smaller size which can be place right on top of the string and be fit to our ukulele holder for position fixing. This part will be complete in the future work.

To provide synchronization among different electronic component we have to connect all of it to one major board for coding. Currently, the sliders and stepper motors are connected to their corresponding driver, and both are wired to an Uno board for in-set communication. Without driver, insufficient power is supplied to the motor, therefore, to provide enough power for rotation. A sensor is also connected to each Uno board. All the I2C ports of four Uno boards are then connected a breadboard hence to the Raspberry Pi for further communication, this enable multiple I2C devices to use the two same pin.

The following is the wiring diagram showing how one set of arm is connected to the Raspberry Pi:

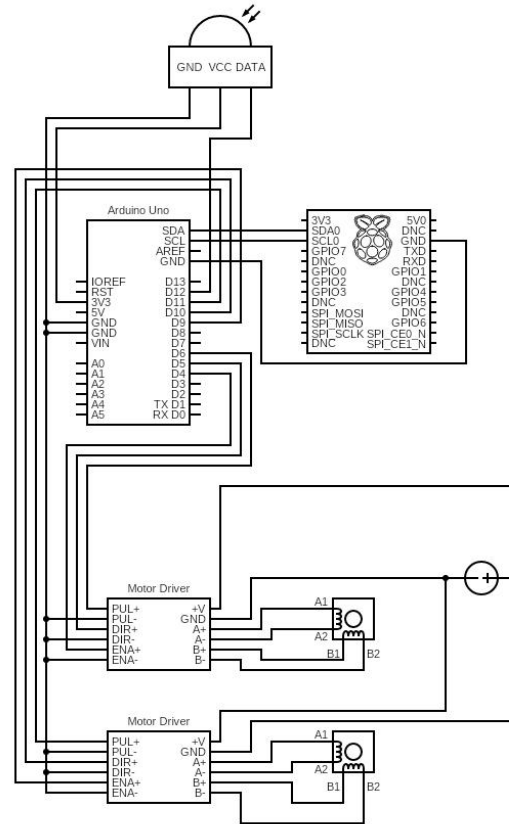


Fig. 9. This graph showcase how the wiring among one set of arms is inter-connected to a Uno board hence a Raspberry Pi. And ultimately there will be four set of Uno board connecting to the same Raspberry Pi.

2) *3D Print*: Within this project, it is necessary to design different components that cater to our needs in terms of functionality and size, for instance, the design of the rack and holder, etc. Among different options, 3D printing technology has been chosen as it can be printed according to our design, providing precision in the required shape and size. The two commonly used materials for prototyping fused deposition modeling (FDM) 3D printing are Acrylonitrile Butadiene Styrene (ABS) and Polylactic Acid (PLA). In the project, it is crucial to have the surface of the 3D print precise and smooth as it minimizes noise and allows different parts to move seamlessly. Therefore, PLA has been chosen due to its higher tensile strength and smoother surface finish [9] .

SOLIDWORK 2024 is being used as a tool to design all the 3D component. With the feature of designing it component by component, it allows us to assemble different part to have a over all

picture of the product and demo movement before printing. This allow problem to be observed and corrective amendment made before printing, reduce the time to amend the draft due to the inability to assemble after printing all components.

Multiple parts are made with 3D printing in this project, namely, the slider stabilizer, ukulele holder, ukulele stand, rack, and a holder placed on the slider to position the rack and motor.

As mentioned, a multi-arm mechanism is tailored to perform the pressing of the strings. There are four sets of arms in total to handle the four corresponding strings. The arm mechanism is composed of two parts: a slider controlling the movement along the x-axis to reach different frets, and a rack with a gear controlled by a stepper motor to perform the movement along the y-axis to press different strings.

The holder is designed to be placed on top of the slider to hold the motor driver, gear, and rack in place. One of the key features of the holder is to leave room for the rack to move while ensuring that the space is not too large, preventing the rack from wobbling during vertical movement. The base of the holder is drawn the same as the plate of the slider, with nine screw holes to allow M5 screws to be installed, ensuring that the holder is tightly secured during movement. Above the base, there is a platform tailored to place our NEMA 17 motor driver. Without the platform, there the screw hole will be covered by the tunnel. Not installing all screws maybe result in error in the position calculation and unwanted wobbling. A leveled platform ensure there is no height difference between the motor and the holder, allowing the screws can be installed and confirming it is leveled. From the front view, a hole is reserved for the shaft to go through. The four screw holes in the front view align with the front endcap of the motor, allowing M3 screws to be installed for positioning.

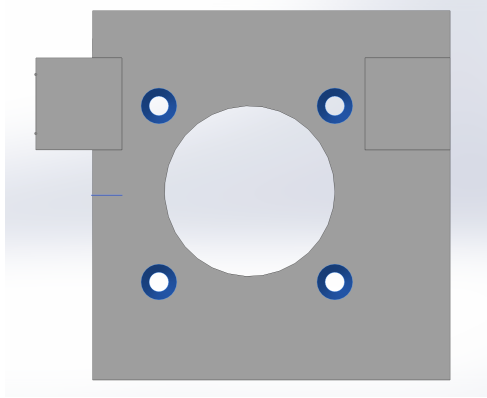


Fig. 10. Front view of holder with four M3 screws.

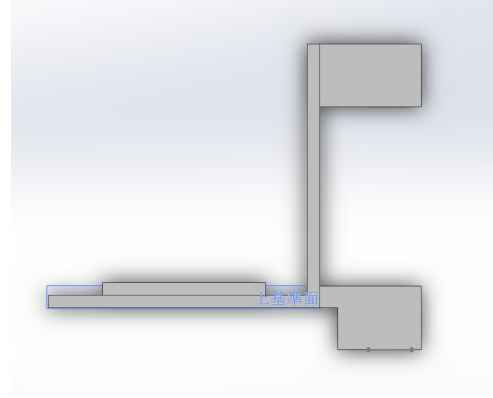


Fig. 11. Side view of holder with platform and two square tunnel featured.

Another feature is the two square tunnels on the side of the holder, which are for placing the rack and guaranteeing that the movement of the rack aligns vertically. Without the tunnel on the side, the rack will not be able to place in the same position and not able to move vertically solely depending of the gear as it may disassemble due to gravity. The entrance of the tunnel is adjusted to be slightly larger than the rack, leaving only minimum space for it to pass through the tunnel and move vertically. Leaving a small gap for movement will still result a strange sound due to wiggling. This problem by adjusting the 3D print to leave minimal space for movement and utilize the action-reaction force applied by the gear in stabilizing the rack. The position of the tunnel is chosen based on the following concern. If the tunnel is too long, it may touches the wooden board and may add constraint to the height of the rack to be higher in order not to clash with the holder. Therefore, we measured the difference of the holder with the board when deciding the length of the lower tunnel and set the upper tunnel length not to exceed the total height of the holder, hence minimize the limitation in affecting the design of the rack.

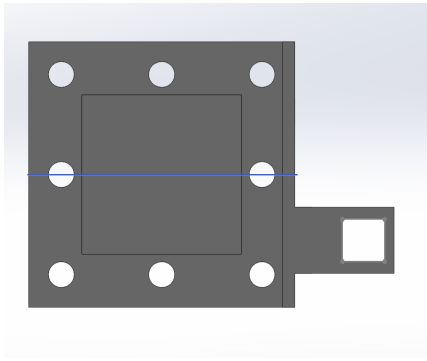


Fig. 12. Top view of holder showcasing the entrance of the tunnel and screw holes of the holder for install screw on the slider.

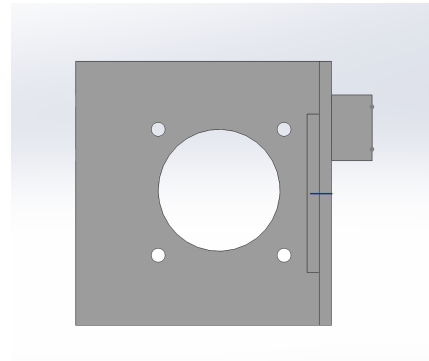


Fig. 13. Back view of holder showing the extended part of the tunnel.

After concerning the height and size of each tunnel, the position of the tunnel carefully designed concerning the mechanism of using gear. First of all we choose the gear with 35 teeth and radius of 1 cm, which is the minimum radius with the maximum teeth we can get. The smaller the gear the less space it take up, while the more teeth the gear have indicating we can have precise control of the length we would like to have. With such approach, the tunnel need to be placed at a position that just fit the horizontal position of the rack when it assemble with the gear and at a position that will not affect the installation of the screws of the motor on the holder. Each slider will be paired with one holder, and the holders for the sliders on the left and right are flipped along the y-axis to match different sides.

The overview of holder draft is provided below:

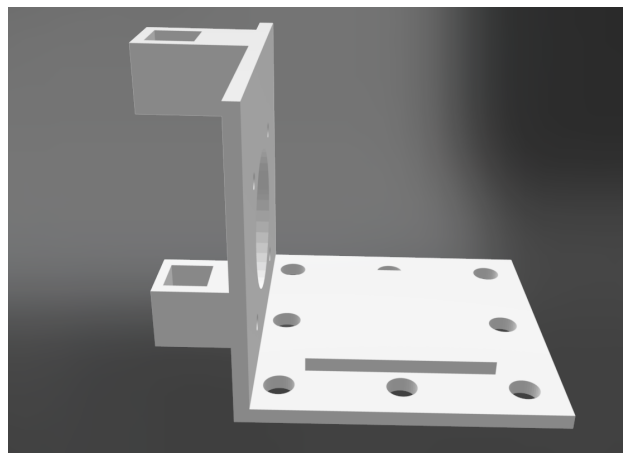


Fig. 14. 3D graph of the holder.

In order to press different strings, a special rack is tailored with two different lengths along the z-axis. The rack is designed to allow precise vertical movement together with the gear, with a series of teeth placed in the middle of the y-axis of the rack. The total length of the series of teeth is measured based on the range of the vertical movement needed for the rack to press the string without clashing. Considering a case that when the inner string is pressed and outer string is not pressed and requires access to another fret. The range of the rack that presses the inner layer should not affect the movement of the outer rack, hence determine the length of the teeth of the rack. The other reason for reserving a recessed area for placing the teeth, with this approach it will not exceed the entrance of the tunnel eliminate all possibility of easily getting stuck in the holder. The other part of the rack, not in contact with the gear, remains smooth to avoid unwanted instability that could occur if the teeth were to get stuck in the tunnel during movement. To press the string, a thin circular stick is added to the tip of the z-axis component of the rack.

The design draft is shown as follows:



Fig. 15. This is the rack designed to place at the left to press the inner string.



Fig. 16. This is the rack designed to place at the right to press the inner string.

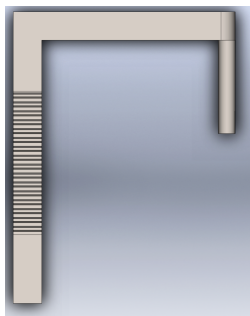


Fig. 17. This is the rack designed to place at the left to press the outer string.

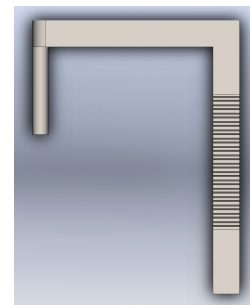


Fig. 18. This is the rack designed to place at the right to press the outer string.

To ensure the multi-arm mechanism can reach the correct position every time, it is necessary to fix both the ukulele and the sliders onto a wooden board. For the ukulele, a holder is designed to hold the body, and a stand is designed to support the neck. The holder is designed to hold the hourglass-shaped body of the ukulele at the thinnest part of the ukulele, near the sound-hole and leaving room for string. By holding the thinnest part of the ukulele can avoid ukulele to move back and forth, meaning the x-axis.

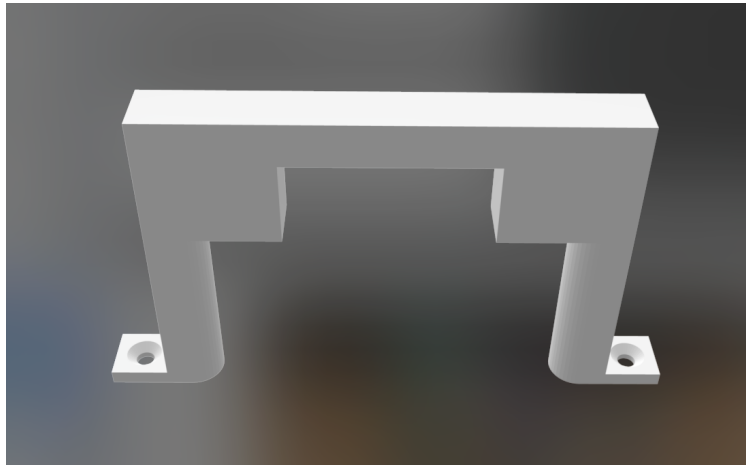


Fig. 19. Ukulele holder with upper room left for strings.

However, the ukulele may move from side to side if it is solely depend of the ukulele holder, to prevent such happened, a ukulele stand is designed to placed at the thinnest part of the neck to secure the movement along z-axis. With the holder pressing down on the body and the stand supporting the neck, a stable and level position is ensured, avoiding unwanted deviations. The design draft is attached below:



Fig. 20. Here is the front view of the ukulele stand.

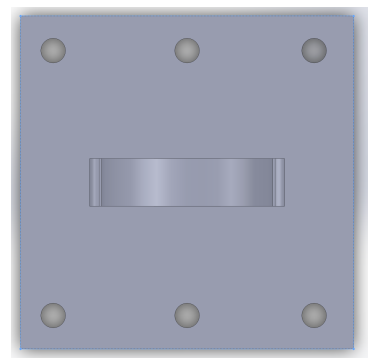


Fig. 21. Here is the top view of the ukulele stand.

Besides the ukulele, stabilizing the slider is also important. The original design of the stand of holder is a triangular shaper stand that support the slider with the tip of the triangular shape. An L-shaped stabilizer is designed to replace all four stands of the original slider. With the screw hole at the bottom part of the L-shaped holder, it allows screws to be installed onto the wooden board. Two separate designs are made to ensure that the stabilizers do not collide with each other.

The design draft is shown below:

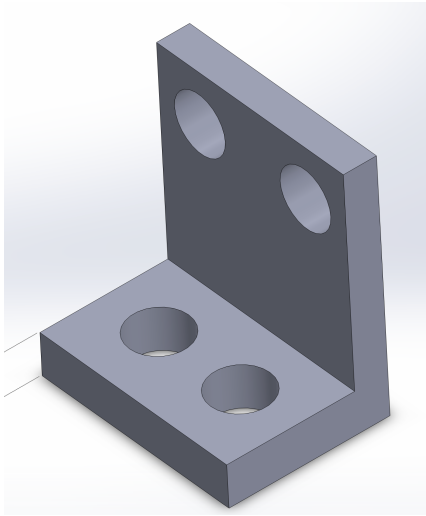


Fig. 22. L-shaped stabilizer.

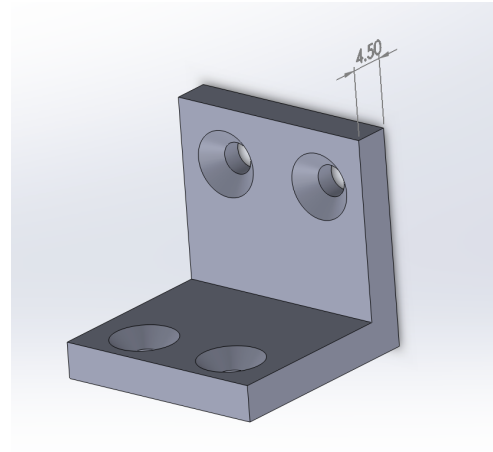


Fig. 23. Shorten version of L-shaped stabilizer to avoid overlapping of components.

B. Software

1) Introduction: The software component consists of three parts: the Arduino Uno, Raspberry Pi, and the mobile application.

For the Arduino Uno, it is used to control the motors. Each Uno board is connected using an I2C interface, with a unique network address assigned to each board. In total, four Uno boards are used—two for controlling the vertical and horizontal movement of the rack, and an additional board for plucking the strings.

The Raspberry Pi is used to receive Bluetooth signals from the mobile device and relay commands to the Arduino Uno boards. It acts as a bridge between the user interface and the hardware components.

For the mobile application, we are using a cross-platform framework. React Native is currently chosen, as it has a large community and supports the simultaneous development of both Android and iOS applications.

2) Mobile Application: The mobile application part is the only part of software that the user will directly interact with. The main feature of the software part is that the user can simply send command to the server side. When the user enters the Application, it will first show a view that the user can select the device they want to interact with. It allows the user to control multiple robots if they have. All device discoverable Bluetooth LE (BLE) device will be shown, while the Guitar Robot will be highlighted.

After that, it will show a connecting view. If the device is not able to connect, it will retry at most 3 times. After that it will show an error to the user. The user can go back to the previous view and try to connect again. When the user is successfully connected. They can input the specified command to the device.

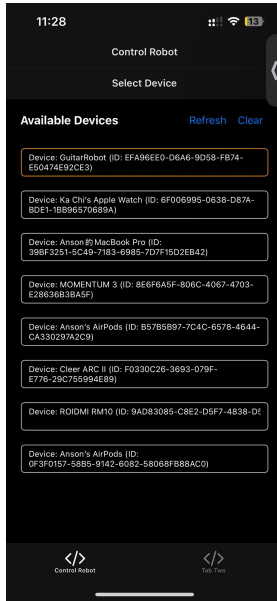


Fig. 24. The user can select the device.



Fig. 25. Connecting View.

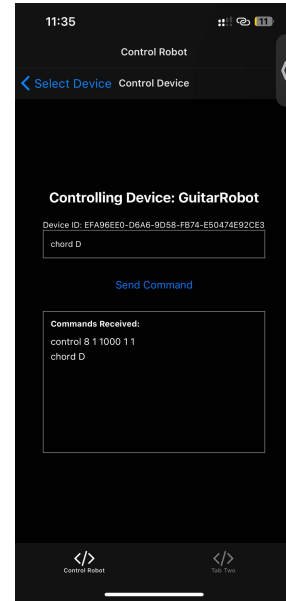


Fig. 26. Enter command View.

The application is written in React Native. We use React Native instead of others like Flutter or native language like swift. First, it has a relative large community where we can find packages and solutions more easily. Second, it uses React like structure and Typescript, which our team is familiar with. Third, it can compile iOS and Android Version at the same time, which reduce the duplicate work on different platforms. Also, we choose Typescript on top of JavaScript, where we can have less runtime errors as we can discover the mismatch of type on compile.

In the application, we have three views currently. They are currently switched by Tabs.Screen and Stack.Navigator. We use Tabs for navigating between device tab (Shown in above figure) and settings tab (Only placeholder in current version). It can preserve the state when we switch between two tabs, to prevent the user need to select device again when they go to another tab. For the control within the device we use a navigator. The user can clearly go back to the device selection page when the are failed to connect to device or wanted to switch to another device.

Also, we have used the package BleManager. It is use to communicate with BLE devices. The reason why we select this package is because it can send the data on a single communication as an array, not bit by bit. We have also considered some BLE Serial Packages, however most of them are outdated and the transmit may be unstable as some bits may be lost. Also, we select

BLE over the Bluetooth Classic is because iOS only supports connecting certified Bluetooth Classic devices on App.

On top of that, we have implemented an Bluetooth service. When we service is first called, it will create a new BleManager Object to the class in the constructor. We placed the BleManager in service instead of UI is because it can prevent create multiple BleManager Object and be interrupted when we switch between different Views. In different UI of program, we can now easily call the BLEManager functions in the service.

When the program is started, the BLE Service will first discover the signal that is currently broadcasting. Also, it will show devices that are already paired and connected. When the user clicks on it, it will try to connect to the device. After that, it will discover it's characteristics, which will mention later in the Raspberry Pi part. If there are characteristics returned, the user will be directed to the command input view. After the user send commands, the program will check if it has a writable characteristics, then send the command to the corresponding universally unique identifier (UUID) in base64 format. If the server lost connection or no corresponding UUID is found, the server will show an error screen and let the user to reconnect again.

3) *Raspberry Pi*: For the Raspberry Pi part, it will first detect all devices reachable in the network connected by the I2C port. Then it will compare them to a dictionary that contains the device information. After confirming all devices are workable, it will start a pairing mode in Bluetooth. The Raspberry Pi automatically accept all incoming pairing requests. We allow multiple device connections because we wish to always keep the device in pairing mode, eliminating the need to remove the pairing every time if we want another device to connect. When the user is connected, they can send commands to the Arduino Uno through the Raspberry Pi.

To setup the device, we have written a script. It will enable the I2C port on Raspberry Pi, which allows the communication to Arduino Uno. Also, it will download all packages for Bluetooth and python. Then, it will modify the settings of the Bluetooth service, including the broadcast name and discovery period etc. After that, it will setup a service to Auto Run the program, hence the user will. User can run the script on a fresh Raspberry Pi firmware and it will install

all dependencies. After reboot, the server should be successfully setup and the user should no longer need to attach the Raspberry Pi to a screen.

When the device is setup, it will automatically run a python program every time on start up. It divides into multiple parts to control different components. We select python as the development language is because it has a large community and there are multiple packages on Bluetooth or I2C connections. After program loads, it will start scanning all I2C devices. After that, it will start the Bluetooth service in a thread. Then, it will start receiving user input.

For the user input part, we will parse the input commands into different components. The user can choose to control the specific motor by selecting the Uno board and the motor target. Also, it can directly input the chord or the calibrate command. The user can input via the terminal or the Bluetooth thread. The terminal is for backup usage in case the user have no Bluetooth device and wants to use the guitar robot. After the command is parsed, it will send to the Uno board by I2C Port.

For the I2C part, it will first compare with the I2C device list that scanned when the program start up when the user tries to send command. If device is not found, it will send an error. Else, it will send the I2C Block data by padding the command into a bit array. The "Control", "Calibrate" and "Move" command is for debug purpose.

	1	2	3	4	5	6	7	8	9
”Control”	0	Target	Speed		Duration				Direction
”Calibrate”	1	Target							
”Move”	2	Target	Distance						
”Fingering”	3	Fingering Details							
”Chord”	4	Chord Details							

For the Bluetooth part, the whole part is run in a thread. It can make the terminal is usable, that the user can input in terminal and via Bluetooth in the same time. It will first initialize the Bluetooth service. It will turn on the Bluetooth LE mode and start broadcasting the device. Also, it will change the IO capabilities, which let the device can be paired without entering a

pin code. It can make the device connectable without a monitor. After the service is initialized, it will start to advertise itself.

To send signal using Bluetooth, we would have two approaches, which is GATT server and Serial to transmit data. However, due to the reason mentioned in Mobile Application part, we can only use GATT as a choice. If GATT service is not enabled, the client side can only pair to the device but cannot transmit data. However, there are no GATT packages for python. We used the dbus interface to interact with the Bluetooth process. After the process is connected, we have attached a MotorService into an Application as a GATT service. After that, the server will advertise its service and characteristic. The service will get value from the client and pass it to the user input part to parse the command.

4) *Arduino Uno*: The Arduino Uno part is written by C language.

First, we would have a test to the motor. It will move the motors in both directions, hence it can ensure the wire are connected.

After that, We will calibrate each motor. For example, we will move the motor to the end to ensure it has the same position every time we start up. We will use an infrared sensor to detect its position. After that, it will turn into discovery mode and let the others send commands to it. It will have a unique network address on I2C that is hardcoded in program. When it receive command, we have also normalized the input format to make instructions more readable, details is included in Raspberry Pi part above.

The Uno board will send digital signal to the start pin and direction pin; also it will send analog PWM signal to the speed pin. Hence, we can control the speed, direction and status of the motor by Uno board. Also, the board will keep tracing how much distance did the motor go forwards and backwards for the chord function.

V. NEXT STEPS

A. *Hardware*

After focusing on the string pressing mechanism, our next goal is to design the sweeping mechanism. We plan to use a servomotor with a blade to mimic the sweeping motion of human fingers when playing chords. The ukulele holder design will be modified to integrate the servomotor into the robot.

Final adjustments will be made to the design of the 3D-printed components to ensure accurate spacing among the different parts, proper positioning of all components, and avoidance of any misalignment. Furthermore, after ensuring the robot's functionality meets our expectations, we plan to rearrange the layout of the robot and modify the appearance of each 3D-printed component to make it more aesthetically appealing.

B. *Software*

To start with, our first target is to enable the guitar robot to play music scores. Currently, our robot is able to play specific chords by sending control commands to the robot. In the next step, we will implement all chords. After that, we plan to implement a score feature that allows the user to input an entire song. This feature will enable the robot to play consecutive notes, forming a complete musical score.

The above idea requires updates from both client application and server side. In the server, it has to be able to receive the score in a customized format. Then, we have to save it in the server side hence it can be called in other devices. In the client side, it can download the scores from server and play it locally; or it can upload the score to the guitar robot. Currently, we now plan to implement it in a JSON like format. We would be able to identify the mode of chord and fingering, different notes and how long does each notes be. To achieve it, we have to update the GATT service of the server side and the Bluetooth method in client side to make it possible. The reason why we would handle the score parsing in server is because we would like to reduce the delay on Bluetooth, as Bluetooth may cause each break between notes uneven.

After that, we would like to implement a score system. That the user can create a score by importing other instruments score like MIDI files, and it can enter the notes by its own editor. The user can preview and play in the editor and finally transfer the score to the robot.

Finally, we may implement an AI system. We would use image recognition to read the score printed on paper. As such, it can transfer to score in our own editor system.

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